
CareerAI

AI-Powered Career Intelligence Platform

INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP

Project: CareerAI — AI-Powered Career Intelligence Platform

SUBMITTED BY:

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Domain: Artificial Intelligence

Date: 08.07.2026

1. Introduction

The project "**CareerAI — AI-Powered Career Intelligence Platform**" is a full-stack web application that functions as an intelligent career coach. It leverages artificial intelligence to analyze resumes, detect skill gaps, predict career paths, and provide real-time AI-powered guidance to help professionals make informed career decisions.

The platform addresses a common challenge faced by job seekers: understanding how their resume performs against applicant tracking systems (ATS), identifying which skills are missing for their target roles, and finding the right career trajectory. CareerAI automates this analysis using NLP-powered resume parsing and large language model inference, delivering actionable insights in seconds.

Users can upload their resume in PDF or DOCX format, receive an ATS compatibility score, view detected skills and skill gaps, explore AI-recommended career paths with salary estimates, browse personalized job and course recommendations, and interact with an AI career coach chatbot that understands their profile context.

2. Objective

The primary objectives of the CareerAI platform are:

- **Resume Analysis & ATS Scoring** — Parse uploaded resumes using NLP techniques, extract structured data (skills, roles, experience), and generate an ATS compatibility score benchmarked against industry standards.
- **Skill Gap Detection** — Identify missing skills relative to the user's target career path and provide prioritized recommendations with explanations for each gap.
- **Career Path Prediction** — Use AI to generate personalized career trajectory recommendations with match percentages, required skills, salary ranges, and timelines.
- **Salary Intelligence** — Provide role-aware, location-aware salary estimates based on the user's detected skillset and experience level.
- **Job & Course Recommendations** — Curate personalized job listings and skill-building courses matched to the user's profile and identified gaps.
- **AI Career Coach** — Offer a conversational chatbot interface that maintains profile context and delivers personalized, actionable career guidance.
- **Secure Authentication** — Implement JWT-based user authentication with registration, login, and profile management.
- **Clean, Professional UI** — Deliver a responsive, accessible interface built with React that prioritizes usability and visual clarity.

3. Tech Stack

The CareerAI platform was built using the following technologies:

Layer	Technology	Purpose
Frontend	React.js + Vite	Single-page application with fast dev server and optimized builds

Layer	Technology	Purpose
Backend	FastAPI (Python)	High-performance async API framework for REST endpoints
Database	MySQL + SQLAlchemy	Relational database with ORM for structured data storage
AI Engine	OpenRouter API (Llama 3.1 8B)	Large language model for resume analysis, career predictions, and chat
Auth	JWT (JSON Web Tokens)	Stateless authentication for secure API access
Resume Parsing	NLP Libraries	PDF/DOCX text extraction and skill entity recognition
Charts	Recharts	Radial and bar charts for ATS score visualization
Icons	Lucide React	Consistent, lightweight icon system
HTTP Client	Axios	Promise-based API communication with request/response interceptors
Version Control	Git & GitHub	Source code management and collaboration

4. System Architecture

The application follows a clean three-tier architecture: a React-based single-page application (SPA) frontend communicates with a FastAPI backend over RESTful JSON APIs, which in turn interacts with a MySQL database through SQLAlchemy ORM. The AI capabilities are provided by an external large language model accessed via the OpenRouter API.

Frontend Layer: Built with React 19 and Vite, the SPA handles routing (react-router-dom), state management (React Context), and API communication (Axios with JWT interceptors). The UI uses a clean design system with CSS custom properties for consistent theming.

Backend Layer: FastAPI serves as the API gateway, handling authentication (JWT token generation and validation), resume file uploads with NLP-based parsing, and AI inference requests. The backend exposes endpoints for auth, users, resumes, jobs, courses, and AI operations.

Data Layer: MySQL stores user accounts, resume data, parsed analysis results, and cached recommendations. SQLAlchemy provides ORM abstraction with automatic table creation on startup.

5. Screenshots and Output

Below are screenshots demonstrating the key features of the CareerAI platform.

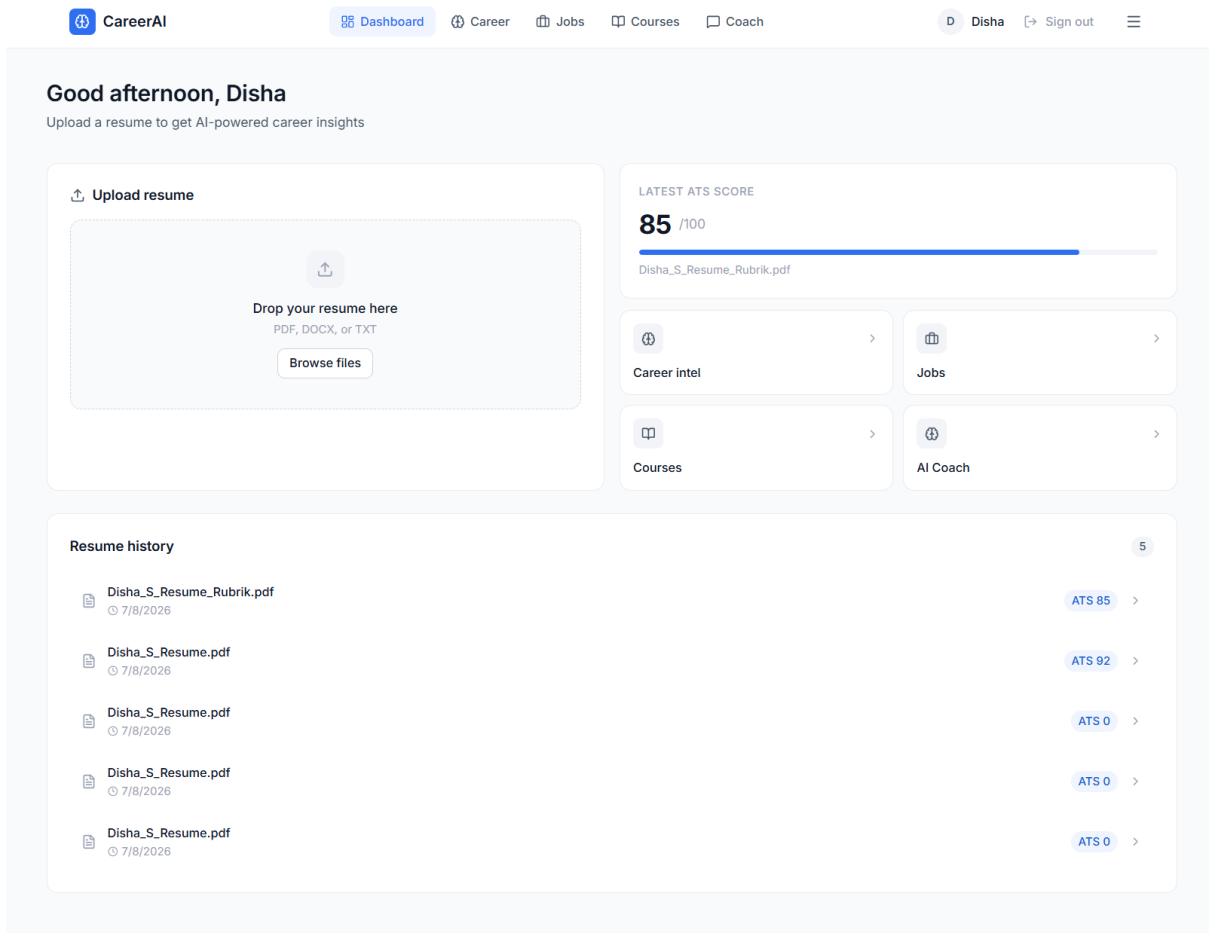


Fig 1: Dashboard — Resume upload, ATS score overview, quick navigation, and resume history

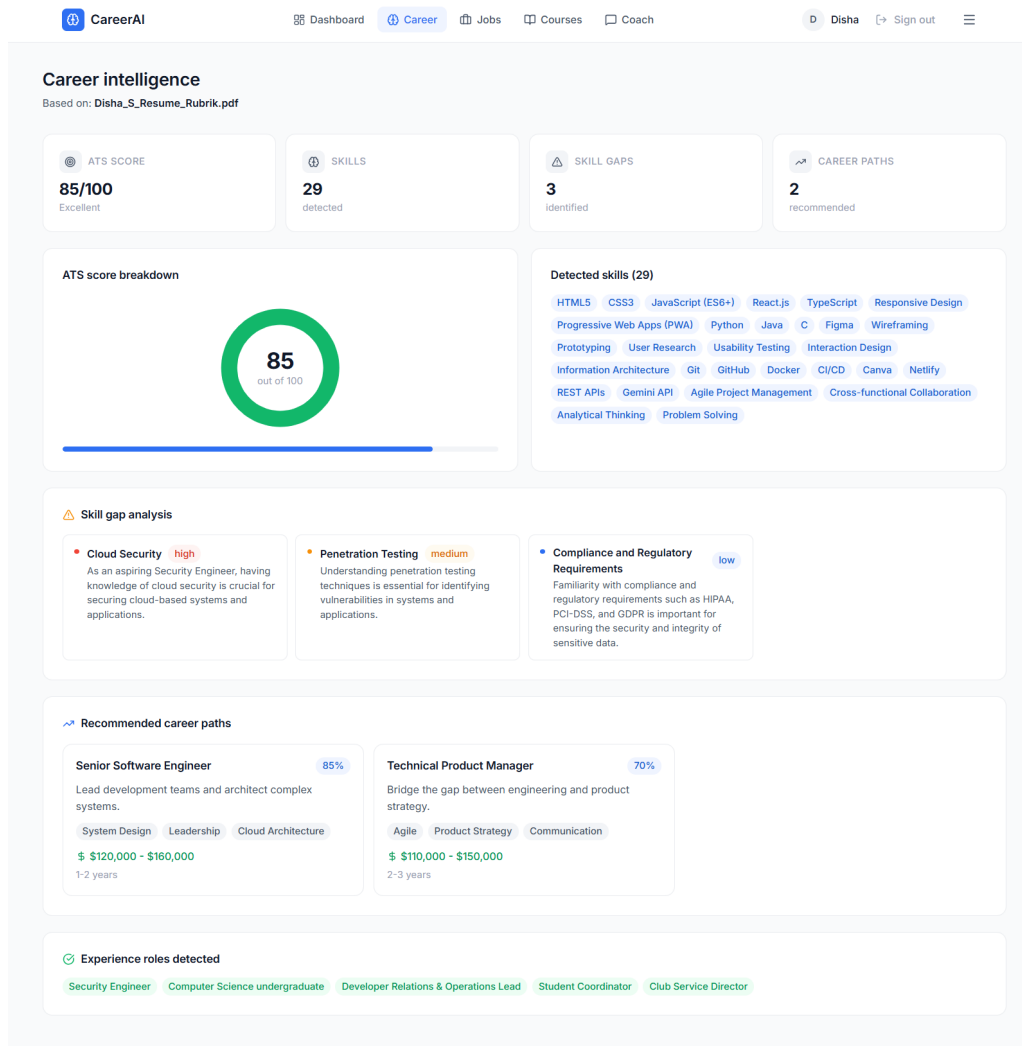


Fig 2: Career Intelligence — ATS score breakdown, detected skills, skill gap analysis, and career paths

CareerAI

DashboardCareerJobsCoursesCoach

D DishaSign out

Jobs

Personalized opportunities based on your skills

Refresh

Filter by title or company...

Showing 5 opportunities

Senior Frontend Developer

Microsoft

92%

Redmond, WA\$ \$120k - \$180kFull-timeSenior

We are seeking a highly skilled Senior Frontend Developer to join our team, responsible for designing and developing responsive, scalable, and maintainable...

React.jsJavaScriptHTML5CSS3Responsive Design

Apply now

Full-Stack Developer

Amazon

88%

Remote\$ \$100k - \$160kRemoteMid

We are looking for a talented Full-Stack Developer to join our team, responsible for designing and developing scalable web applications using Python, Java, and...

PythonJavaReact.jsJavaScriptHTML/CSS

Apply now

Software Engineer - UI/UX

Google

95%

Mountain View, CA\$ \$140k - \$200kFull-timeSenior

We are seeking a highly skilled Software Engineer to join our UI/UX team, responsible for designing and developing intuitive and responsive user interfaces...

FigmaWireframingFrontend DevelopmentUI/UX DesignJavaScript

Apply now

AI/ML Engineer

Facebook

90%

Menlo Park, CA\$ \$160k - \$220kFull-timeSenior

We are looking for a talented AI/ML Engineer to join our team, responsible for designing and developing scalable machine learning models using Python, Java, and C.

PythonJavaCMachine LearningAI

Apply now

Frontend Developer

Airbnb

85%

San Francisco, CA\$ \$90k - \$140kFull-timeMid

We are seeking a highly skilled Frontend Developer to join our team, responsible for designing and developing responsive and scalable web applications using...

React.jsJavaScriptHTML5CSS3Responsive Design

Apply now

Fig 3: Job Recommendations — AI-curated job listings with match scores and skill tags

CareerAI

DashboardCareerJobsCoursesCoach

D DishaSign out

Courses

Learning resources to close your skill gaps

Refresh

Filter by title or provider...

Showing 6 courses

Cloud Architecture with AWS

Udemy

85%

12 hours4.7Intermediate

Learn to design and deploy scalable cloud architectures using AWS...

Cloud Architecture

Start learning

Machine Learning with Python

Coursera

90%

4 weeks4.5Beginner

Learn the basics of machine learning using Python and popular libraries like...

Machine LearningPython

Start learning

DevOps Essentials

LinkedIn Learning

80%

6 hours4.3Beginner

Learn the fundamentals of DevOps, including continuous integration,...

DevOps

Start learning

Cloud Native Applications with Kubernetes

edX

88%

8 weeks4.6Intermediate

Learn to design and deploy cloud-native applications using Kubernetes....

Cloud ArchitectureDevOps

Start learning

Machine Learning with TensorFlow

YouTube

82%

10 hours4.5Intermediate

Free

Learn to build machine learning models using TensorFlow. This course covers...

Machine Learning

Start learning

DevOps with AWS

freeCodeCamp

92%

12 hours4.8Intermediate

Free

Learn to implement DevOps practices using AWS services like CodePipeline,...

DevOpsCloud Architecture

Start learning

Fig 4: Course Recommendations — Skill-building courses matched to identified gaps

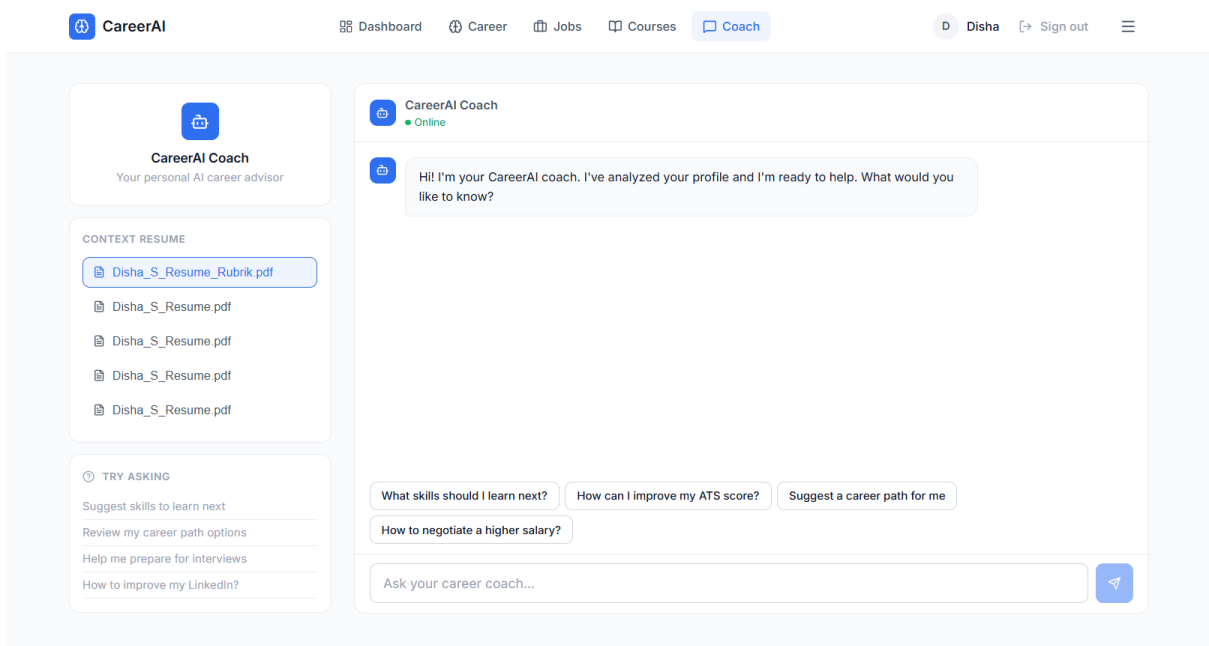


Fig 5: AI Career Coach — Conversational interface with resume context and quick prompts

6. Key Features Summary

Resume Upload & Parsing

Supports PDF, DOCX, and TXT formats. Uses NLP to extract skills, roles, experience, and education from unstructured resume text.

ATS Score Analysis

Generates a compatibility score (0–100) against applicant tracking system benchmarks, displayed with a radial chart and progress bar.

Skill Gap Detection

Compares detected skills against target role requirements. Each gap includes a priority level (high/medium/low) and an explanation.

Career Path Prediction

Recommends career trajectories with match percentages, required skills, salary ranges, and estimated timelines.

Job Recommendations

AI-curated job listings with match scores, company info, location, salary, job type, and direct apply links.

Course Recommendations

Personalized learning resources targeting skill gaps, with match scores, ratings, duration, and provider info.

AI Career Coach

Chatbot interface with resume context awareness, quick prompt suggestions, and conversation history.

Responsive Design

Clean, professional UI that works across desktop and mobile viewports with a consistent design system.

7. Conclusion

The CareerAI platform successfully delivers an integrated, AI-powered career intelligence system that addresses the key challenges job seekers face today. By combining resume parsing, ATS scoring, skill gap analysis, career path prediction, and an interactive AI coach into a single platform, it provides a comprehensive toolkit for career development.

The project demonstrates the practical application of large language models in career services, showing how AI can transform unstructured resume data into structured, actionable insights. The clean architecture — separating concerns across React frontend, FastAPI backend, and MySQL database — ensures the system is maintainable and scalable.

Future enhancements could include integration with live job boards via APIs, LinkedIn profile analysis, multi-language resume support, interview preparation modules, and collaborative features for career mentors and coaches.

8. Acknowledgment

I would like to express my sincere gratitude to the Infosys Springboard team and my mentor for their continuous guidance and support throughout the development of this project. This virtual internship has been a valuable experience in learning full-stack development, AI integration, and building real-world applications that solve meaningful problems.

The mentorship and structured project framework provided by the Infosys Springboard program were instrumental in shaping the direction and quality of this project. I am grateful for the opportunity to work on a project that combines artificial intelligence with practical career development tools.